Department of Computer Science & Engineering
UNIVERSITY OF LIBERAL ARTS
BANGLADESH

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Course Title: Database Management System Lab

Section: 02

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Open-Ended Lab

Submitted to:

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Project Title: Daily Expense Management System

1. Introduction

The importance of personal finance management has greatly increased at this point in time. Many people have problems monitoring their expenditures and end-up making bad decisions about how to spend their money. Personal Expense Tracker System is an application based on a database that helps monitor, record and evaluate expenditures.

This open-ended experiment focuses on developing a comprehensive Personal Expense Tracker System that enables user to:

- Register and maintain personal profiles
- Add, edit, and delete expense records
- Categorize expenses for better organization
- Generate Reports based on different time periods
- Visualize spending patterns through charts and graphs

The system follows an architecture consisting of a MySQL database backend, PHP business logic layer, and HTML/CSS frontend interface. This report documents have the complete development process with requirement analysis, ER diagram design, database implementation, application development, and testing.

2. Problem Requirements

2.1 Problem Statement:

Create a database-based Personal Expense Tracker System that can store, update, delete, and retrieve data regarding expenses efficiently. It should enable the user to manage his/her daily expenses and create reports based on such data for financial analysis in a proper way.

2.2 Functional Requirements:

Requirements	Description
1. User Registration	Users can create an account using name, email, and password

2. User Authentication	Registered users can securely log into the system
3. Add Expense	Users can add new expenses including amount,date, and category
4. View Expenses	Users can view all recorded expenses in a structured and sorted format
5. Edit Expenses	Users can modify existing expense record
6. Delete expense	Users can remove unwanted or incorrect expense entries
7. Category Management	System provides predefined categories such as Food, Transport etc.
8. Dashboard	Users can view recapped statistics (daily, weekly, monthly, yearly, total)
9. Reports	System generates expense report for daily,weekly,mothly and yearly basis.
10. Profile Management	Users can update their personal information and profile picture
11. Password Management	Users can securely change their account password

2.3 Non-Functional Requirements:

Requirement	Description
1. security	User passwords are securely stored using MD5 hashing
2. Data Integrity	Database relationships are enforced using foreign key constraints
3. Usability	System provides a simple and user-friendly interface
4. Performance	Optimized queries ensure for the fast retrieval of expense reports and data
5. Portability	The system is compatible with standard XAMPP server environments

2.4 Future Improvements

Potential enhancement for future versions:

1. Budget planning and alerts
2. Receipt/image upload for expenses
3. Export Reports to PDF/Excel
4. Multi-currency Support
5. Data backup and restore functionality

3. Objectives

The primary objectives of this open ended experiment are followed as:

1. To analyze the requirements of a personal expense tracking system and identify key entities and relationships.
2. To design a database model and shows on Entity-Relationship (ER) diagram.
3. To implement the database schema using MySQL.
4. To develop a complete web-based application using PHP and MySQLthat supports create,read, update and delete operations.
5. To create meaningful reports and visualizations from the stored expense data.
6. To demonstrate database normalization techniques (3NF compliance).
7. To document the complete development process in a structured lab report format.

4. Modern Tools

- Luchidchart - Diagram creation
- Visual Studio Code (VS Code) - Coding
- Web Browser - Visualization tools
- MySQL (XAMPP) - Database management
- Windows - Operating System

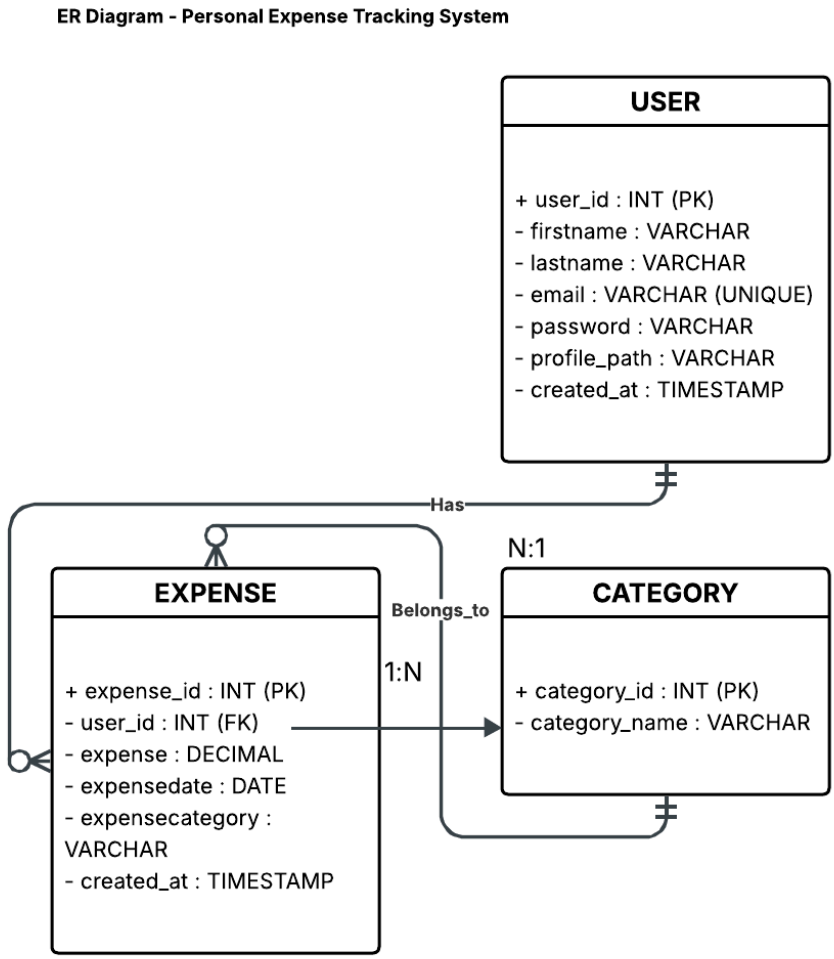
5. Background Study

For this project, I carried out background research on how the Personal Expense Tracker System assists people in managing their financial transactions in a systematic manner. While traditional approaches like recording transactions manually are time-consuming, a computerized system becomes necessary for accurate and efficient results. All the data regarding user details, expenditure, and categories is recorded in an orderly fashion so as to ensure easy and effective analysis of information. Every transaction recorded in this system belongs to a particular user and category. It was found that security during logging in into the system plays a crucial role to keep the data safe. There are automated calculations

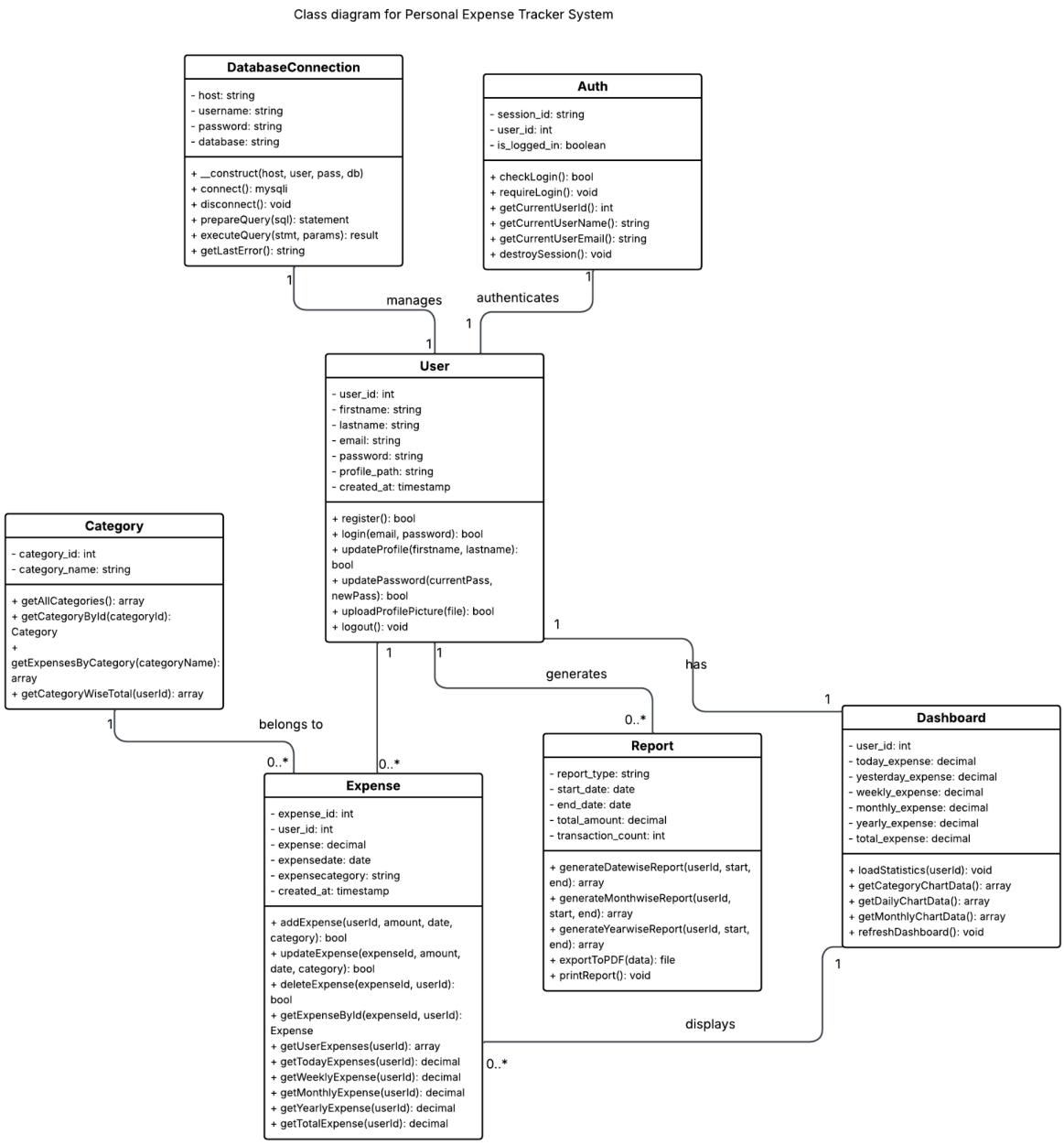
performed by this system that give the sum total of expenses incurred over certain periods such as per day or month.

6. Diagram:

1. ER diagram:



2. Class Diagram:



6.1 System Outputs:

Personal Expense Tracker System generates various outputs depending on user actions such as logging into the system, creating a new account, adding expenses, and generating reports. Once a user logs in or creates an account successfully, a message will appear notifying him of his success, and he will be directed to the dashboard. Should the user enter wrong details while trying to log into the system, an error message will be displayed. In case of a successful expense addition, the system updates its database and generates the output list of all expenses.

6.2 Explanations:

This system works using a database where user information, expenses, and categories are stored. Every expense is associated with a particular user and particular categories to

facilitate tracking. This system is designed in such a way that it provides information on the total cost in certain time frames, such as per day, per week, or per month. Moreover, this system also creates various reports.

6.3 Screenshots:

The following are screenshots which reflect the live interface of the Personal Expense Tracker System showing how various modules work in practice.

Figure 1: Login Screen:

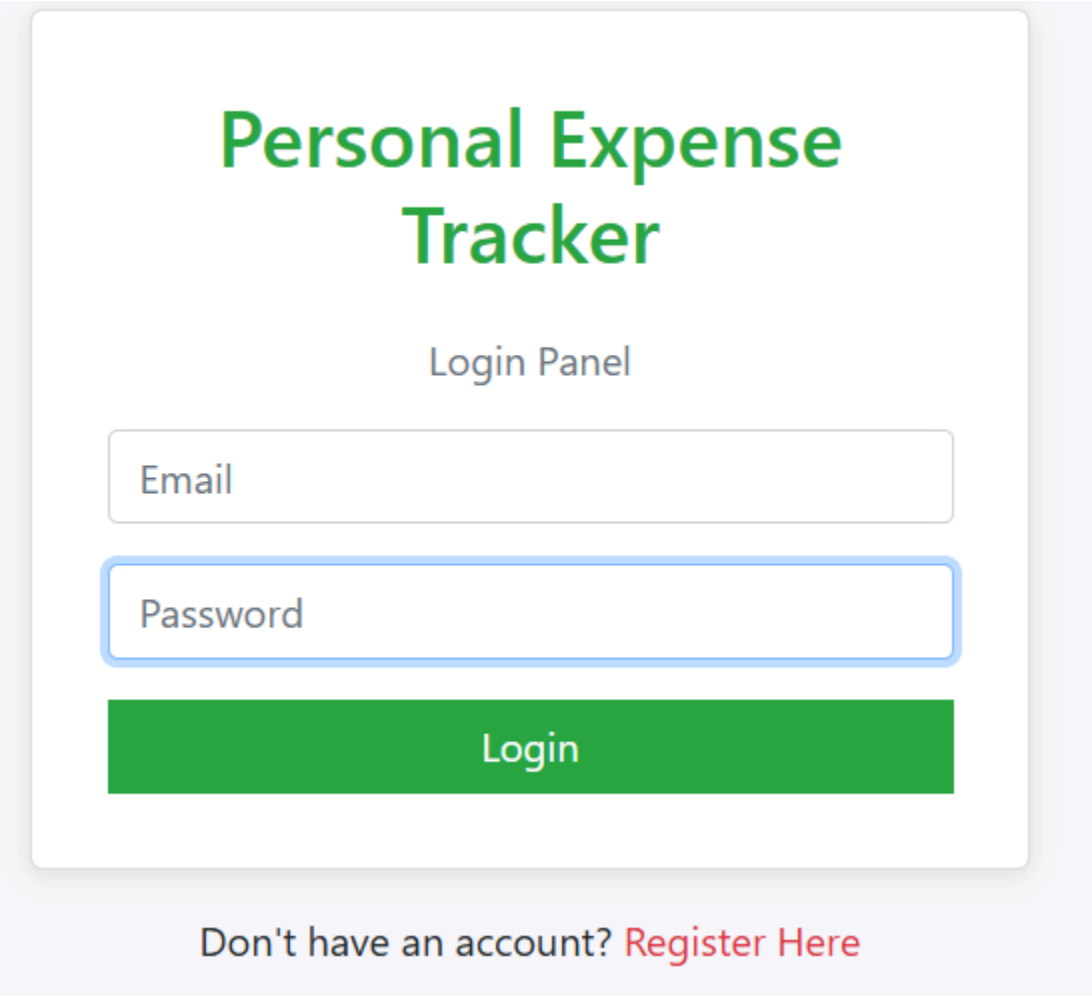


Figure 2: Register Screen:

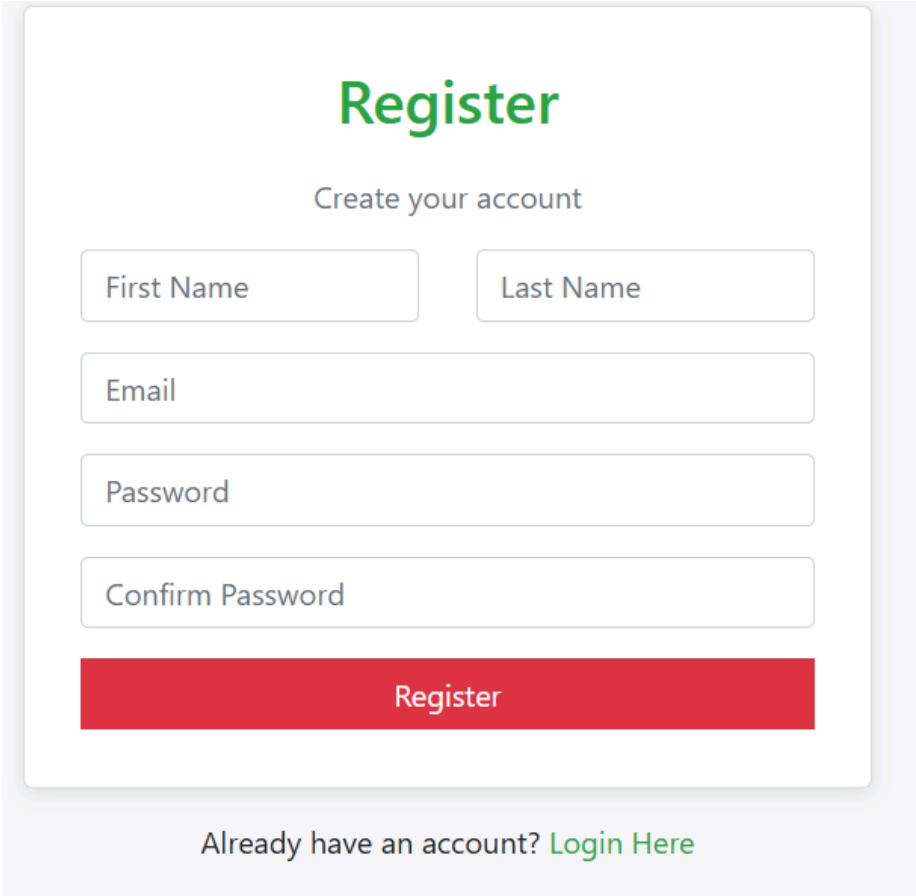


Figure 3: Dashboard Screen:

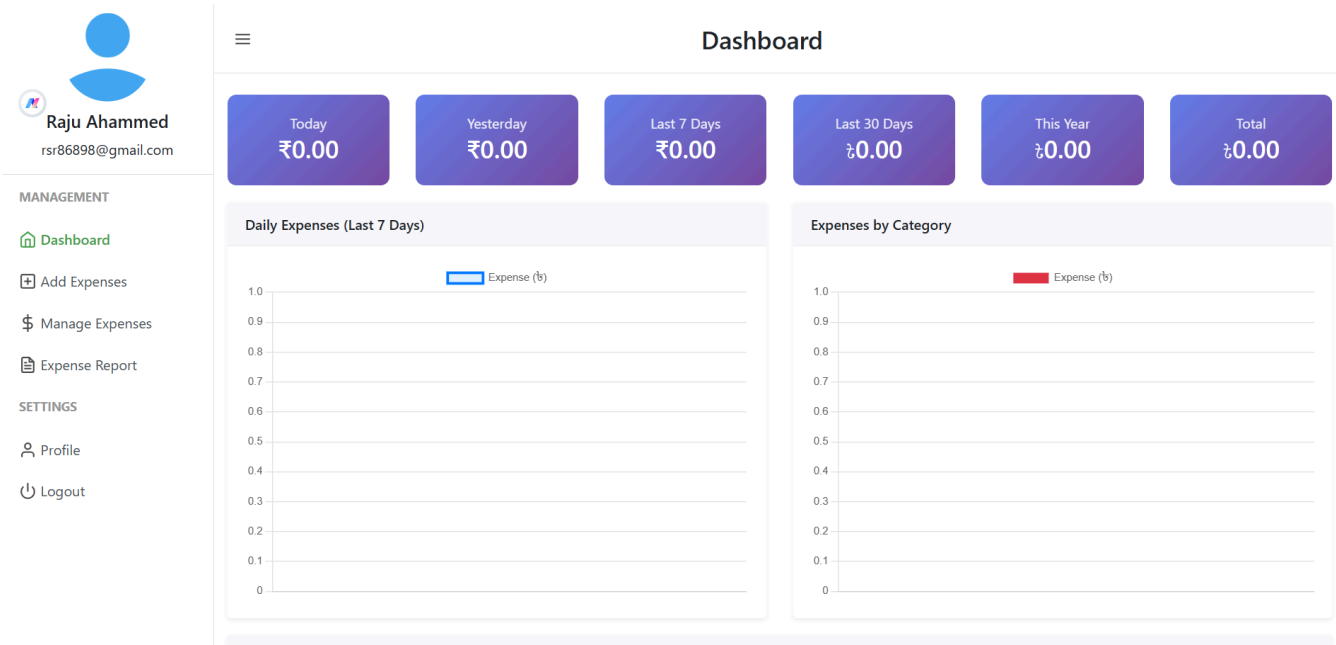


Figure 4: Add Expense Screen:

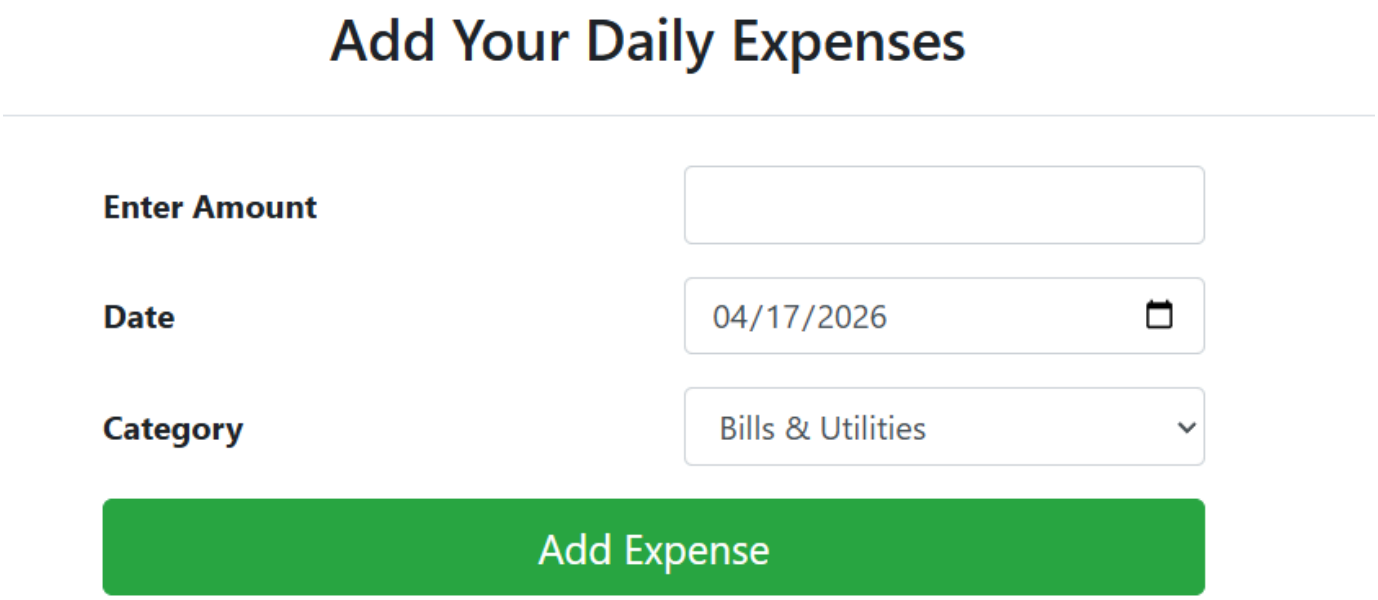
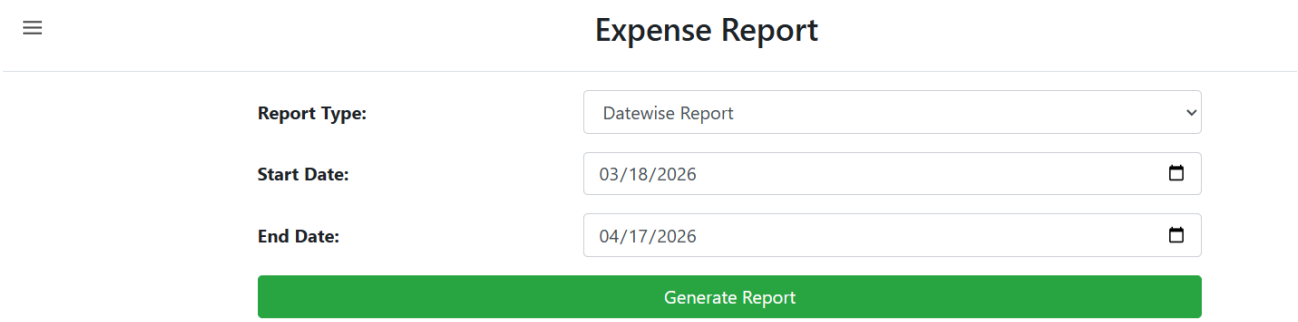
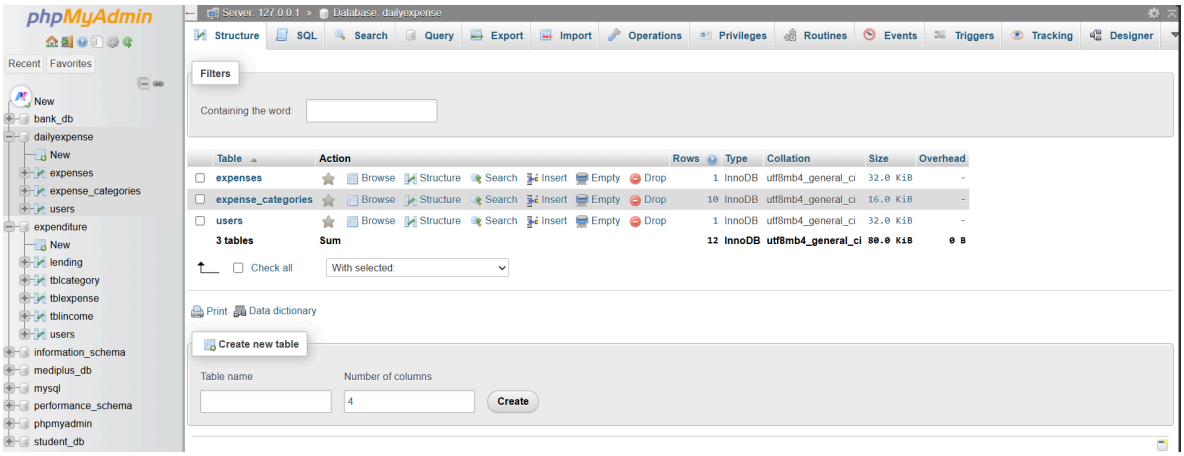


Figure 5: Expense Reports Screen:



Create Database and Tables:



Create database dailyexpense if its not exist. Then create table (table name: expenses, expense_categories, users) then insert values into the expenses_categories.

```sql
-- Database: `dailyexpense`
--
CREATE DATABASE IF NOT EXISTS `dailyexpense` DEFAULT CHARACTER SET utf8mb4 COLLATE utf8m
USE `dailyexpense`;

-- --------------------------------------------------------

--
-- Table structure for table `expenses`
--

CREATE TABLE `expenses` (
  `expense_id` int(11) NOT NULL AUTO_INCREMENT,
  `user_id` int(11) NOT NULL,
  `expense` decimal(10,2) NOT NULL,
  `expensedate` date NOT NULL,
  `expensecategory` varchar(50) NOT NULL,
  `created_at` timestamp NOT NULL DEFAULT current_timestamp(),
  PRIMARY KEY (`expense_id`),
  KEY `user_id` (`user_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

```sql
-- Table structure for table `expense_categories`
--

CREATE TABLE `expense_categories` (
  `category_id` int(11) NOT NULL AUTO_INCREMENT,
  `category_name` varchar(50) NOT NULL,
  PRIMARY KEY (`category_id`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

--
-- Dumping data for table `expense_categories`
--

INSERT INTO `expense_categories` (`category_id`, `category_name`) VALUES
(1, 'Food'),
(2, 'Transport'),
(3, 'Shopping'),
(4, 'Entertainment'),
(5, 'Bills & Utilities'),
(6, 'Healthcare'),
(7, 'Education'),
(8, 'Rent'),
(9, 'Travel'),
(10, 'Others');
```

```sql
-- Table structure for table `users`
--

CREATE TABLE `users` (
  `user_id` int(11) NOT NULL AUTO_INCREMENT,
  `firstname` varchar(50) NOT NULL,
  `lastname` varchar(50) NOT NULL,
  `email` varchar(100) NOT NULL,
  `password` varchar(50) NOT NULL,
  `profile_path` varchar(255) DEFAULT 'default_profile.png',
  `created_at` timestamp NOT NULL DEFAULT current_timestamp(),
  PRIMARY KEY (`user_id`),
  UNIQUE KEY `email` (`email`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;
```

8. Conclusion

The Personal Expense Tracker System is an example of the successful implementation of the theories related to the database management system covered in class. It allows the user to store, update, delete, and retrieve expense data, in addition to generating useful information from reports and visualization tools. In this way, the system meets all the requirements of the open experiment.

9. References:

1. MySQL Documentation. Learned from [https://dev.mysql.com/doc/]

2. W3Schools. PHP and MySQL Tutorials.[https://www.w3schools.com/]

3. Laboratory Manual and Practical Session Instructions.

4. YouTube Educational Videos. PHP, MySQL, and Web Development Tutorials. Learned

from [https://www.youtube.com/]